SUBRAHMANYESWAR KOLLURU

AI Engineer | Machine Learning Engineer | Computer Vision | LLM Systems

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PROFESSIONAL SUMMARY

Published AI researcher and B.Tech (AI & ML) undergraduate with hands-on experience building production-grade AI systems across computer vision, LLM applications, edge inference, and AI automation. Designed and deployed a TensorRT-optimized edge AI pipeline achieving 447 FPS at 2.23ms latency on standard CPU hardware. Published at ICCCNT 2025 (IIT Indore). Actively building multi-agent systems, advanced RAG pipelines, and local LLM deployments. Seeking AI Engineer / ML Engineer roles for 2026–2027 placements.

TECHNICAL SKILLS

Languages	Python, C++ (Basics), SQL
AI / ML	TensorFlow, Keras, PyTorch, Scikit-Learn, OpenCV, YOLO, ArcFace, MobileNetV2, LSTM, Transfer Learning, Transformers
LLM & GenAI	LangChain, LlamaIndex, Ollama, Llama 3, OpenAI Whisper, RAG Pipelines, Vector Databases, Embeddings, Unsloth, HuggingFace, Fine-Tuning, Transformers
AI Agents	CrewAI, Multi-Agent Systems, MCP Protocol, Agent Orchestration, Tool Integration
Deployment	FastAPI, TensorRT, Intel OpenVINO, FP16 Quantization, Edge AI, Docker (Basics), Streamlit, Next.js, Redis, WebSocket
Cloud	AWS (S3, EC2, RDS, Lambda, IAM), MongoDB Atlas
Data & DB	MongoDB, Neo4j, MYSQL, NumPy, Pandas, Matplotlib
CS Concepts	Data Structures & Algorithms (Intermediate), OOP
Dev Tools	VS Code, Jupyter Notebook, Git, GitHub, linux

WORK EXPERIENCE

AI Engineering Intern | READ Automation Pvt. Ltd., Chennai — *Nov 2025 – Jan 2026*

Project: github.com/Subrahmanyeswar/Bottle-Cap-Inspection-System

- **447 FPS at 2.23ms latency — 23.8% faster than TensorFlow Lite baseline, running on standard laptop CPU.**Engineered a 4-stage Edge AI pipeline (Video Acquisition → Motion Trigger → ROI Extraction → MobileNetV2 Inference) achieving
- **95% classification accuracy and 100% recall on critical Missing Cap defects.**Applied transfer learning on MobileNetV2 (ImageNet weights, ~200 custom images + augmentation); achieved
- **48% (8.9 MB → 4.6 MB) and CPU utilization by 15.1%; implemented motion-based triggering cutting idle overhead by 60%.**Optimized inference via Intel OpenVINO FP16 quantization — reduced model size by
- Applied CLAHE preprocessing (LAB color space, clip limit 3.0, 8×8 tile grid) ensuring >92% accuracy across 200–2000 lux factory lighting. Research paper published on this work.

PROJECTS

Aeternum Hub — GPU-Accelerated AI Security System | *Python, YOLO11m, TensorRT, ArcFace, ViT-GPT2, FastAPI, Redis, Next.js*

GitHub: github.com/Subrahmanyeswar/aeternum-hub

- **<100ms detection latency at 25–30 FPS with 3–5x GPU speedup; ArcFace achieving 95%+ face recognition accuracy with <2% false positive rate.**Architected a full-stack AI security platform — YOLO11m (TensorRT) achieving
- **100% local processing — DPDP Act 2023 compliant, zero cloud dependency.**Built 3-stage progressive alert system (5s/10s/20s) with SMS + voice calls + auto video recording; ViT-GPT2 pipeline for AI-generated police reports.
- Published at ICCCNT 2025, IIT Indore | Top 10 National Project Expo (100+ teams) | 3rd Prize national AI/ML Hackathon.

Multi-Agent Competitive Intelligence Platform | *CrewAI, Python, LLMs, Multi-Agent Systems*

GitHub: github.com/Subrahmanyeswar/multi-agent-competitive-intelligence

- Designed a CrewAI multi-agent orchestration pipeline with 3 specialized agents (Research, Analysis, Synthesizer) for autonomous competitive landscape monitoring with parallel information processing and structured weekly report generation.

Advanced RAG System & LLM Infrastructure | *LangChain, LlamaIndex, Vector DB, Ollama, HuggingFace, LoRA/PEFT*

GitHub: github.com/Subrahmanyeswar/Aeternum-AI-Advanced-RAG

- Built an end-to-end RAG pipeline with document ingestion, chunking, embedding generation, and vector similarity search for context-aware, grounded LLM responses with reduced hallucinations.
- Deployed local LLM inference via Ollama (offline, zero API cost); conducted LoRA/PEFT fine-tuning experiments on open-source LLMs using HuggingFace for domain-specific task performance.

Jarvis AI Assistant | *Python, Groq API, Meta LLaMA 3, YOLOv8, OpenCV, Speech-to-Text, TTS*

GitHub: github.com/Subrahmanyeswar/Jarvis-AI-Assistant

- Built a modular voice-controlled AI assistant using Groq API (Meta LLaMA 3) for NLP, speech-to-text + TTS for hands-free interaction, YOLOv8 + face recognition for intruder detection, and full PC automation via voice across 5 independent modules.

EDUCATION

B.Tech — Computer Science & Engineering (AI & ML) — *CGPA: 8.27 | 2023 – 2027*

Kalasalingam Academy of Research and Education (KARE)

ACHIEVEMENTS & CERTIFICATIONS

- Research Publication — ICCCNT 2025, IIT Indore (AI Security)
- National Project Expo — Top 10 finalist (100+ teams)
- Hackathon — 3rd Prize, national-level AI/ML competition